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[58/27N]

SARDAR PATEL UNIVERSITY

M.Sc (IT) Semester-I Examination – November 2023 [PS01CINT54] : Relational Database Management System

2nd November 2023 (Thursday)

Time: 10.00am To 1.00pm

Total Marks: 70

Note: Figure indicates right is maximum marks for each question

Q-1 Select an appropriate option for the following statements [08]

- 1 _____ are logical structures used to organize and store data for future processing, retrieval, or evaluation.
[a] Databases [b] Table [c] Rows [d] Columns
- 2 _____ is at the lowest level tier in three-level architecture and responsible for managing and storing data.
[a] Physical [b] Conceptual [c] External [d] None of these
- 3 A piece of data that can be uniquely identified in DBMS is known as _____.
[a] Relationship [b] Entity [c] Attribute [d] Tuple
- 4 When one or more items in one table can have a relationship to one or more items in another table is called _____ type of relationship.
[a] one-to-one [b] one-to-many [c] many-to-many [d] none of these
- 5 Which command is used to remove the records with structure from table?
[a] Delete [b] Drop [c] Truncate [d] All of these
- 6 _____ is a type of join that returns the Cartesian product of rows from the tables in the Join.
[a] Self Join [b] Inner Join [c] Cross Join [d] Outer Join
- 7 _____ cursor attribute returns number of records in a cursor.
[a] %Count [b] %Recordcount [c] %Rowcount [d] None of these
- 8 From the following which PL/SQL section is used to handle errors?
[a] Exception [b] Error [c] ErrorHandler [d] None of these

Q-2 Answer the following questions in brief. (Attempt Any SEVEN) [14]

- 1 What is the need of RDBMS?
- 2 Explain Tuple and Attributes.
- 3 What is E-R Diagram? List its components.
- 4 What is Recursive Relationship in DBMS?
- 5 Explain entity with its types.
- 6 Explain MOD() and SUBSTR() function with syntax and example.
- 7 Discuss basic DML commands with its use.
- 8 Compare SQL with PL/SQL.
- 9 What are the basic cursor attributes? Explain any one of them in brief.

[P.T.O]

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Q-3

[A] What is Data Model? List categories of data models and discuss any Two data model in detail. [06]

[B] What do you mean by Data Sharing and Data Protection? Discuss with suitable example. [06]

OR

[B] Discuss about Roles played by different users associated with DBMS.

Q-4

[A] What is Normalization? Discuss different forms of normalization with example and its needs. [06]

[B] Write a note on Codd's rule. [06]

OR

[B] Discuss Relationship Cardinality with example.

Q-5

[A] Assume **Employee_Master** table and write sql command to perform following task. [06]
Employee_Master(Emp_id , Emp_name, Birth_date, Department, Salary, City)

1. Arrange name of employees in ascending order.
2. Display the details of employees whose salary is between 40000 and 70000.
3. List name of employees whose name starts with letter 'M'.
4. Change the city of employee to 'Mumbai' whose Emp_id is 120.
5. Add a new column Date_of_joining(date) to table Employee_Master:
6. Display employee details who belongs to Ahmedabad and working in IT department.

[B] What is Join? List its types and discuss any two of them with suitable example. [06]

OR

[B] Explain Integrity Constraints in detail.

Q-6

[A] Discuss Exception Handling with its use. [06]

[B] Explain structure of PL/SQL Block with each of its section. Also discuss advantages of PL/SQL over SQL. [06]

OR

[B] Write a PL/SQL block to update salary for a given Emp_id in Employee_Master table. Use following table.

Employee_Master(Emp_id , Emp_name, Birth_date, Department, Salary, City)
